

ANMOL BARUWAL

Nashville, TN | 413-273-9005 | anmolbaruwal01@gmail.com | [GitHub](#) | [LinkedIn](#)

SUMMARY

AI Engineer with ~1 year of production experience building and orchestrating multi-agent systems. Shipped three production agents (invoicing, underwriting, marketing) and ControlPanel, a FastAPI orchestration platform with guardrails, state-machine runs, and a Slack/API front end, cutting ~10 hrs/week of manual work. Strong in Python, FastAPI, the Anthropic/OpenAI SDKs, RAG (Pgvector + hybrid retrieval), and production reliability patterns.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C, C++

AI Tools: Multi-agent orchestration, MCP tool calls, Policy Gate, A2A protocol, Anthropic/OpenAI SDKs, RAG, Pgvector, BM25, hybrid retrieval (RRF), confidence gating, JSON-schema validation, agentic workflows

Backend & Infra: FastAPI, Pydantic, SQLAlchemy (async), PostgreSQL, Supabase, Docker, Railway, AWS (Lambda, S3), Kafka, CI/CD (Jenkins, GitHub Actions)

Web & Frontend: React, Next.js, Tailwind CSS, HTML5, SCSS

EXPERIENCE

Elite Capital Group - Nashville, TN

Mar 2026 to Present

AI Engineer

- Building a production multi-agent orchestrator in Python and FastAPI that classifies inbound requests, plans tool calls across a registry of domain agents; every run and step persists to PostgreSQL behind an explicit state machine, so execution state is recoverable and auditable rather than in-memory.
- Designed a deny-by-default policy layer gating every agent action through an allowlist of 17 typed operations with strict argument schemas and append-only audit events, and enforced a no-autonomous-send invariant by static analysis over the source tree, making an unreviewed send structurally impossible rather than prompt-dependent.
- Built an offline evaluation harness for routing and response generation that scores each case over repeated trials and passes only on unanimous agreement, run as a pre-deploy regression gate; took the suite from 39/45 to 45/45 while fixing a class of misrouted requests.
- Shipped a document-extraction agent pairing vision-model reading with deterministic parsing fallback, confidence scoring, range validation, and HITL for ambiguous records, removing ~10 hrs/week of manual invoice entry.

Five Fold University (Registered 501(c)(3) Non-Profit) - Nashville, TN

May 2025 to Mar 2026

Full-Stack Developer

- Architected and maintained a multi-tenant CMS and LMS educational platform (React, multi-role auth, relational data model) serving a global student body across 12+ countries.
- Shipped responsive student, instructor, and admin UIs with event instrumentation for per-cohort engagement tracking, engineering a secure system from database schema through API to UI.

Vanderbilt University - Nashville, TN

Feb 2025 to Apr 2025

Software Engineering Intern

- Automated IT incident management across TeamDynamix, Statuspage, and Microsoft Teams APIs on an iPaaS platform serving 10,000+ daily campus users, cutting manual update effort ~30% per incident.
- Engineered an API-driven REST deduplication service within the integration layer to merge related incoming telemetry alerts, providing on-call engineers with a single consolidated outage incident.
- Revamped primary IT infrastructure sites leveraging HTML, JavaScript, and responsive design systems to improve user engagement metrics by 45% and ensure structural alignment with university branding updates.

Apple - San Diego, CA

May 2024 to Aug 2024

Software Engineering Intern (Audio Products)

- Designed the Minimum Marketing Version aggregation logic for AirPods charging-case firmware in C/C++ on an RTOS, reverse-engineering Apple Accessory Protocol (AAP) struct definitions and update payloads to collapse independently versioned cores into one consistent version string.
- Audited firmware structs across bud firmware and the iOS receiver to identify a variable-length member memory hazard; re-specified the implementation to safely reuse an unassigned reserved byte, preserving full backward compatibility.
- Enforced mandatory Jenkins validation on every firmware change to eliminate a critical CI build-bypass loophole, unblocking 200+ firmware engineers, cutting build times by 5%, and recovering ~90 engineer-hours/week.

PROJECTS

Conversation Infra

Jan 2026 to Present

- Built a stateful multi-agent system with conversation routing, persistent history, rate limiting, and parallel agent workers.
- Implemented hybrid retrieval (Pgvector HNSW + BM25 + Reciprocal Rank Fusion) with an async embedding pipeline, delivering real-time matches via Supabase triggers.

TechBridge Learning

Aug 2024 to Present

- Built a learning platform with a FERPA/COPPA-compliant RAG over 68+ curated sources, plus multi-role auth and instructor analytics.
- Tuned vector search (IVFFlat) for sub-100ms queries over 1,000+ chunks; shipped with end-to-end test coverage (Vitest, Playwright) and CI/CD on Vercel.

EDUCATION

Fisk University - Nashville, TN

May 2025

B.S. Computer Science